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Physical understanding of keyhole and weld pool dynamics in laser welding under different water pressures

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ABSTRACT

Underwater laser welding, an efficient and labor-saving in-situ repair technology, is prospective in the field of underwater environment manufacturing. It simplifies the tedious process of disassembly and transportation when the underwater facilities are usually repaired ashore. However, the keyhole and weld pool dynamics in laser welding under different water pressure are little understood. In this study, we developed a mathematical model of underwater laser welding process with arbitrary ambient pressures for the first time. The underlying physics of the underwater laser welding process was explored, and the keyhole and weld pool dynamics between the underwater environment and the ground were compared. Results showed that a keyhole similar to that in the atmosphere is also formed in the water environment. And the keyhole was oscillated. The surface of the weld pool in contact with water cooled quickly, forming a special morphology of the weld pool. The flow in the weld pool was mainly driven by the recoil pressure. The temperature of the surface of the keyhole increased with the increase of water depth. That is because the laser beam energy density is large enough to heat the local metal to the boiling point which increases with the ambient pressure increase. It means that the weld pool was able to absorb more energy before evaporation. Then, less energy is applied to induce the recoil pressure to deepen the keyhole. As a result, a shallower but more stable keyhole is formed in underwater laser welding process under 5 MPa pressure. This study can provide better understanding of the heat and mass transfer behaviors in underwater laser welding.

1. Introduction

Underwater laser welding is a prospective technology for efficient repair of the underwater equipment, such as nuclear power equipment and offshore oil platform [1–4]. With the rapid development of fiber laser and semiconductor laser [5,6], the great increase of laser power (over 100,000 Watts) [7], and the miniaturization of laser device [8], the underwater laser welding method becomes feasible in water environment. In addition, the laser can be transmitted with great accuracy through optical fiber. That makes the laser welding / repairing technology accessible in deep water region [9,10]. Besides, the laser welding technology can produce small heat affected zone and residual stress, which can effectively guarantee the welding quality [11,12]. Therefore, the underwater laser welding technology has an important potential in welding / repairing the large scale structures at considerable depths in the future. However, due to the effects of water environment make the welding process more complicated, the welding mechanisms of underwater laser welding remains unclear.

The study of underwater laser welding began in 1980s. The concept of wet underwater laser welding was first put forward by Sepold and Teske [13]. They observed the physical process of underwater laser welding and found that the plasma can significantly influence the laser beam energy transferred to the workpiece. In 1994, Shannon et al. studied the wet underwater laser welding of 50D steel by using 1.2 kW CO_2 laser [14]. They found that the attenuation of the laser beam was a function of the lens and water depth. In 2006, Zhang et al. studied the relationship between optical signal and welding stability of local dry underwater laser welding process [10]. In 2017, Xing et al. found that the weld depth could be over 20 mm when the laser power was 6.0 kW [3]. BIAS Company of Germany, IHI, Toshiba, and Hitachi of Japan have developed several kinds of shielding devices to maintain the local dry region. Those devices have been used in laser welding of shallow water environment and have been applied to the maintenance of nuclear power facilities. However, laser welding in deep water is still difficult and lacking due to the unfeasible

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experiment. In deep water, the ambient pressure is more than 5 MPa, resulting in complicated interactions of water-laser-metal [15–17]. The fluid flow and heat transfer in the weld pool are changes, influencing the weld quality directly [4,11,18,19]. For example, the lower water temperature in deep water will lead to rapid cooling of weld seam, which will produce brittle hard microstructure and affect the weld performance. Although the preliminary applications of underwater laser welding have been realized, the behaviors of heat transfer and fluid flow in weld pool are lack of understanding, especially in deep water.

In this study, a mathematical model of laser welding process in water under different water depth was proposed to study the dynamic behaviors of the keyhole and the weld pool. The physical behaviors in underwater laser welding process were reproduced and the keyhole and weld pool dynamics under the underwater environment and the ground were compared. The influence of the water depth, represented by the ambient pressure, was also analyzed. This study can provide theoretical understanding of the heat and mass transfer behaviors in underwater laser welding.

2. Materials and methods

The physics of the actual laser welding in the water is complex due to the interactions of physics, chemistry, biology and so on [20–22]. It is a difficult task to integrate the whole physics of the process into a mathematical model. Several simplifications of underwater laser welding process should be made in order to make the model tractable. Fig. 1 shows the diagram of our underwater laser welding model. A local dry laser welding process was investigated. The workpiece was placed under water. The shield gas was used to maintain a local dry environment near the top surface of the workpiece. The laser beam propagated in the gas and irradiated the workpiece surface. In the dry region, we considered the shielding gas pressure on the weld pool surface without considering dynamical behaviors of the shielding gas and vapor mixture for efficient computation. In the wet region, we ignored the influence of the water flow on the weld pool, and treated the strong cooling effect of the water on the metal surface as a forced convection according to some studies of metal quenching [23,24]. Size of the local dry region was determined according to the experiment setup. In addition, considering the laser beam propagated in the dry region, the water attenuating effect on laser was not need calculating and the laser absorption by the weld pool surface was taking into account based on Fresnel absorption formula.

2.1. Governing equations

In order to simulate the heat transfer and fluid flow inside weld pool in underwater laser welding process, we adopted a three dimensional transient laser welding model [25] by assuming that the molten metal liquid is incompressible [26]:

$$\nabla \cdot \vec{U} = 0, \tag{1}$$

$$\rho(\frac{\partial \vec{U}}{\partial t} + (\vec{U} \cdot \nabla)\vec{U}) = \nabla \cdot (\mu \nabla \vec{U}) - \nabla p - \frac{\mu}{K}\vec{U} - \rho \vec{g} \beta (T - T_{ref}), \tag{2}$$

$$\rho C_p(\frac{\partial T}{\partial t} + (\vec{U} \cdot \nabla)T) = \nabla \cdot (k \nabla T), \tag{3}$$

where μ_l is the is the kinematic viscosity of the fluid, ρ is the density, p is the pressure, $\vec{g}$ is the three-dimensional gravitational acceleration vector, β is the thermal expansion coefficient, T_{ref} is the reference temperature, and K is the Carman-Kozeny coefficient in the mixed-phase model. C_p is the specific heat and k is the thermal conductivity [27,28].

The keyhole free surface evolutions can be tracked with Level Set method and the time dependent keyhole profiles can be described as [29]

$$\frac{\partial \phi}{\partial t} + \vec{U} \cdot \nabla \phi = 0, \tag{4}$$

where ϕ is a signed distance. The transient free surface of the keyhole can be described as the zero contour the distance field ϕ.

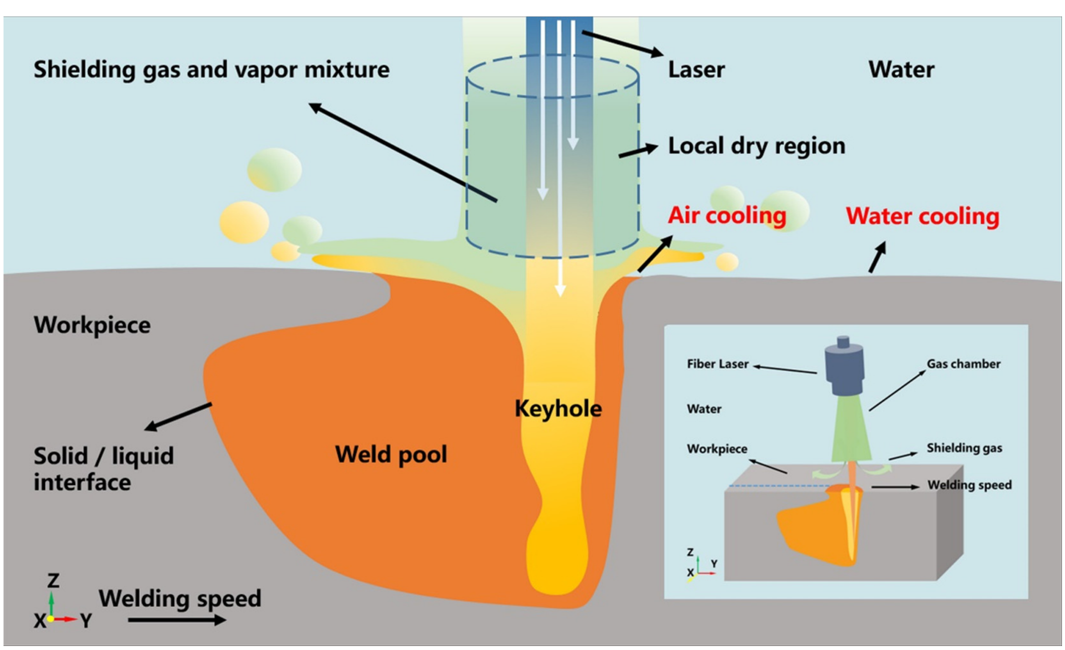

Fig. 1. The diagram of underwater laser welding model.

2.2. Boundary conditions

For the momentum boundary conditions of the keyhole and weld pool, important physical factors, including the Marangoni force, hydraulic pressure, the recoil pressure and the surface tension, are considered together with the sharp interface method developed by Pang [30–32]. On the normal direction of keyhole and weld pool surface, the coupled recoil pressure, surface tension and fluid pressure satisfied a pressure boundary condition:

$$p = p_s + \sigma\kappa + 2\mu_l \vec{n} \cdot \nabla\vec{U} \cdot \vec{n}, \tag{5}$$

where σ is the surface tension coefficient, is the recoil pressure, κ is the curvature, μ_l is the density of molten metal, and p_s is the surface pressure.

For the laser welding underwater, the pressure on the weld pool surface contains shield gas pressure and recoil pressure induced by evaporation. To maintain a stable local dry environment, the gas pressure should be equal to the hydraulic pressure. For taking the shield gas pressure into account, we adopted the improved recoil pressure model, which considered the effect of the ambient pressure on the weld pool surface in Fig. 2 [33–35]. The improved recoil pressure model can be expressed as

$$p_s(T_s) = \begin{cases} p_{amb} & 0 \leqslant T_s < T_L \\ \frac{1+\beta_R}{2} p_0 \exp\left(\frac{\Delta H_v}{k_B T_v}\left(1 - \frac{T_v}{T_s}\right)\right) & +\infty > T_s \geqslant T_R \\ p_c(T_s) & T_L \leqslant T_s < T_R \end{cases}, \tag{6}$$

where p_{amb} is a given ambient pressure, k_B is Boltzmann constant, β_0 is condensation coefficient, T_v is the boiling temperature under the atmospheric pressure p_0, ΔH_v is the enthalpy of phase transition during vaporization, and $p_c(T_s)$ within the range of temperature (T_L, T_R) around is a smooth curve described by a cubic polynomial as

$$p_c(T_s) = aT_s^3 + bT_s^2 + cT_s + d, \tag{7}$$

where *a, b, c,* and *d* are the coefficients related to the ambient pressure. The ambient pressure can be determined according to the water depth.

On the tangential direction of the free surface, the viscous stress momentum boundary conditions due to the Marangoni forces and viscous forces of the molten metal can be described as:

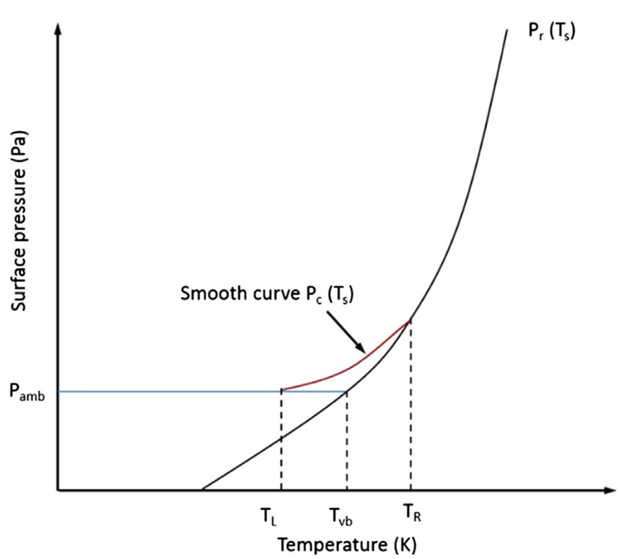

Fig. 2. The schematic of the surface pressure model dependence with surface temperature.

$$\begin{aligned}(\mu\nabla\vec{U})_f &= \mu(\vec{n}\ \vec{t}_1\ \vec{t}_2)(\vec{n}\ \vec{0}\ \vec{0})^T(\nabla\vec{U})(\vec{n}\ \vec{0}\ \vec{0})(\vec{n}\ \vec{t}_1\ \vec{t}_2)^T \\ &+ \mu(\vec{n}\ \vec{t}_1\ \vec{t}_2)(\vec{0}\ \vec{t}_1\ \vec{t}_2)^T(\nabla\vec{U}) - \mu(\vec{n}\ \vec{t}_1\ \vec{t}_2)(\vec{n}\ \vec{0}\ \vec{0})^T \\ &\times(\nabla\vec{U})(\vec{n}\ \vec{0}\ \vec{0})(\vec{n}\ \vec{t}_1\ \vec{t}_2)^T \\ &+ (\vec{n}\ \vec{t}_1\ \vec{t}_2)\begin{bmatrix} 0 & \nabla_s\sigma\cdot\vec{t_1} & \nabla_s\sigma\cdot\vec{t_2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}(\vec{n}\ \vec{t}_1\ \vec{t}_2)^T,\end{aligned} \tag{8}$$

where $\vec{t}_1$ and $\vec{t}_2$ are the normal tangent vectors perpendicular to $\vec{n}$. ∇_s is the surface gradient.

$$k\frac{\partial T}{\partial \vec{n}} = \begin{cases} q_{laser} - h(T - T_\infty) - q_{evp}, & \textit{in local dry region} \\ -h(T - T_\infty) - q_{evp}, & \textit{in wet region} \end{cases} \tag{9}$$

where q_{evp} is the heat loss due to evaporation, h is the heat convection coefficient, T_∞ is water temperature, and q_{laser} is the heat flux from laser beam irradiation. For considering cooling effect of the water, strong forced convection on part of the metal surface contacting water was applied, as shown in Fig. 1. Air convection was applied on the surface locating in the local dry domain. In the present work, the local dry region was cylindrical. Its axis was same with the laser beam axis. Its radius was chosen to be 1.5 times of the weld width for taking the water cooling effect into consideration and avoiding the huge calculation burden from too large local dry region.

Using the ray tracing method and Fresnel absorption formula, the absorbed laser beam energy q_{laser} on the metal surface can be expressed as [31]:

$$q_{laser} = I(r,z)(\vec{I}\cdot\vec{n})\alpha_{Fr}(\theta) \tag{10}$$

$$\alpha_{Fr}(\theta) = 1 - \frac{1}{2}\left(\frac{1+(1-\varepsilon\cos\theta)^2}{1+(1+\varepsilon\cos\theta)^2} + \frac{\varepsilon^2 - 2\varepsilon\cos\theta + 2\cos^2\theta}{\varepsilon^2 + 2\varepsilon\cos\theta + 2\cos^2\theta}\right), \tag{11}$$

where $I(r,z)$ is the beam energy density distribution, $\vec{I}$ the unit vector of incident beams, $\vec{n}$ the unit normal vector of the point on the metal surface, θ the angle between the incident beam and the normal vector, α_{Fr} the Fresnel absorption coefficient, ε a coefficient related to the types of lasers. $I(r,z)$ is modeled as a Gaussian function [31]:

$$I(r,z) = 3P/\left(\pi R^2\right)\exp\left(-3\left(r^2\right)/R^2\right) \tag{12}$$

where R is the effective beam radius, and P is the laser power.

For the thermal boundary conditions on other surfaces of the plate, water cooling and evaporation heat loss were considered. When the plate was penetrated, the water cooling was replaced by the air convection on part of the bottom surface. The boundary condition was

$$k\frac{\partial T}{\partial \vec{n}} = -h(T - T_\infty) - q_{evp}. \tag{13}$$

2.3. Numerical implementations

The simulations were carried out with an in-house software, which was developed with the C++ programming language. The discretization of the equations and the solution were described detailed in [36,37]. The finite difference method was used to discrete the equations. The time- discretization of governing equations, result in

$$\nabla\cdot\vec{U}^{n+1} = 0, \tag{14}$$

$$\rho^n \frac{\overrightarrow{U}^{n+1} - \overrightarrow{U}^n}{\Delta t} + \tilde{C}(\overrightarrow{U}^n) = \tilde{D}(\mu^n \nabla \overrightarrow{U}^n) - \nabla p^{n+1} + \tilde{K}(\overrightarrow{U}^n) + \tilde{F}^n, \quad (15)$$

$$\rho^n C_p^n \frac{T^{n+1} - T^n}{\Delta t} + \tilde{C}(T^n) = \tilde{D}(kT^{n+1}), \quad (16)$$

where $\tilde{C}(\overrightarrow{U}^n)$, $\tilde{D}(\mu^n \nabla \overrightarrow{U}^n)$,$\tilde{K}(\overrightarrow{U}^n)$ and $\tilde{F}^n$ represent the convective term, diffusive term, Darcy term and buoyancy term, respectively. The fifth order WENO scheme and second order central difference scheme are used to discrete the convective term and the diffusive term. With projection method, decoupling of $\overrightarrow{U}^{n+1}$ and p^{n+1} was carried out:

$$\rho^n \frac{\overrightarrow{U}^{*} - \overrightarrow{U}^n}{\Delta t} + \tilde{C}(\overrightarrow{U}^n) = \tilde{D}(\mu^n \nabla \overrightarrow{U}^n) + \tilde{K}(\overrightarrow{U}^n) + \tilde{F}^n, \quad (17)$$

$$\rho^n \frac{\overrightarrow{U}^{n+1} - \overrightarrow{U}^{*}}{\Delta t} = \nabla p^{n+1}, \quad (18)$$

where $\overrightarrow{U}^{*}$ is the temporary velocity. Taking the divergence on both sides of the Eq. (18), one can obtain the pressure Poisson equation.

$$\frac{\rho^n}{\Delta t} \nabla \cdot \overrightarrow{U}^{*} = \nabla^2 p^{n+1}. \quad (19)$$

After using the central difference scheme to discrete the Poisson equation, a PCG (Pre-conditioned Conjugate Gradient) method was adopted to solve the linear equations. The Level Set equation was solved with a high order explicit method following the method presented in reference [37]. The detailed calculation flow chart is shown in Fig. 3. The OpenMP language was used for parallel computation.

For the underwater laser welding of 304 stainless steel, the values of the coefficients calculated for the improved recoil pressure model are shown in Table 1, where the boiling temperatures under different pressure were calculated according to Clausius-Clapeyron equation. The physical parameters used in the simulations are shown in Table 2 [29]. Although surface tension and thermal-capillary coefficients force tend to change by the oxidation, the extent of oxidation was hard to be determined and the coefficients were used without considering the oxidation in Table 2.

3. Results and discussion

3.1. Verification of the accuracy of simulation results

In general, it is very difficult to carry out the underwater laser welding experiment directly in the ocean or lake. The physical simulation experiments in the laboratory are feasible, and have been studied in laboratory for years [1–4,8–19]. To validate the developed underwater laser welding model, the predicated weld dimensions were compared with the experimental data [38]. Kawahito et al. developed a set of complex and precise devices for the underwater laser welding experiments [38]. For clarity, some points of the experiments were referred as follows. The pressure vessel was filled with water, and the pressure can be adjusted to simulate the water pressure at some depth simply. The maximum output power of the fiber laser was 3 kW. The laser wavelength was 1030 nm. The focus spot diameter was 0.1 mm and the focal length of the lens was 200 mm. The shielding gas (N_2) flux was set to 80 L/min. The water temperature was about 13–15 degrees Celsius. In the experiments, 6 mm thick 304 stainless steel was used as the welding material. The process parameter is shown in Table 3.

Fig. 4(a) showed the comparison between the experimental weld cross section and the simulation results of underwater laser welding under 1 MPa. It is easy to know that the weld section was of narrow columnar structure, which indicated the welding mode was the keyhole mode [39,40]. The weld morphology of the simulation result was consistent with the experimental. The simulation result was a bit deeper and thinner. The differences may be caused by the simplified consideration of the environmental effects and the neglect of the effect of the gas in the keyhole. We also compared the simulated of weld depth and the experimental results under atmosphere pressure, 0.25 MPa, 0.5 MPa and 0.75 MPa underwater pressure respectively, as shown in Fig. 4 (b). The simulation results were in agreement with the experimental. Some differences occurred because the model did not consider the water directly but consider the pressure and the cooling effect

Table 1
The coefficients of surface pressure model for laser welding of 304 stainless steel under variable ambient pressure.

Ambient pressure (MPa)	1	5
T_{vb}	3710	4324
T_R	3600	4200
T_L	3900	4500
a	0.02	−2.47
b	−180.14	31.76
c	6.47e5	−2.67e5
d	−7.73e8	5.65e8

Table 2
Physical parameters used in the simulation.

Physical parameters	Units	Value
Density ρ	$kg \cdot m^{-3}$	7200
Thermal conductivity coefficient k	$W \cdot m^{-1} \cdot K^{-1}$	35.0 (297 K)
Specific heat C_p	$J \cdot kg^{-1} \cdot K^{-1}$	760 (297 K)
Kinematic viscosity μ	$kg \cdot m^{-1} \cdot s^{-1}$	0.006
Latent heat of fusion C_f	$J \cdot kg^{-1}$	6.0e5
Latent heat of evaporation C_v	$J \cdot kg^{-1}$	6.52e6
Solidus temperature T_s	K	1727
Liquids temperature T_l	K	1697
Evaporation temperature T_v	K	3100 (Under 0.1 MPa)) 3317(Under 0.25 MPa) 3502 (Under 0.5 MPa) 3621 (Under 0.75 MPa) 3710 (Under 1.0 MPa) 4324 (Under 5.0 MPa)
Surface tension coefficient σ	$N \cdot m^{-1}$	1.0
Thermal-capillary force coefficient σ_T	$N \cdot m^{-1} \cdot K^{-1}$	−0.43e−3

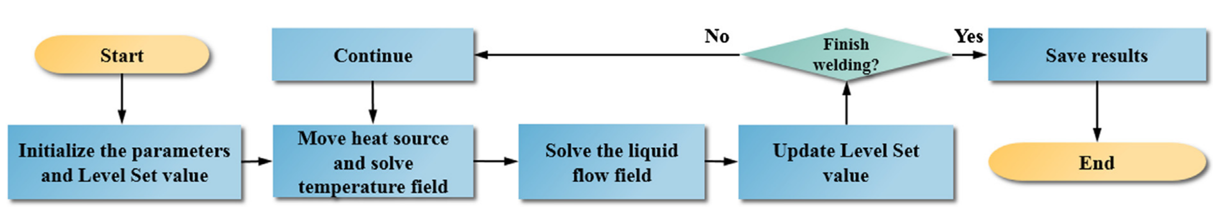

Fig. 3. The simulation flow chart of underwater laser welding.

Table 3
The investigated underwater laser welding parameters [38].

Process no.	Environment pressure (MPa)	Laser power (kW)	Welding speed (m/min)	Defocusing distance (mm)	Environment
1	0.1	3	1.0	0	Atmosphere
2	0.1	3	1.0	0	Underwater
3	0.25	3	1.0	0	Underwater
4	0.5	3	1.0	0	Underwater
5	0.75	3	1.0	0	Underwater
6	1	3	1.0	0	Underwater
7	5	3	1.0	0	Underwater

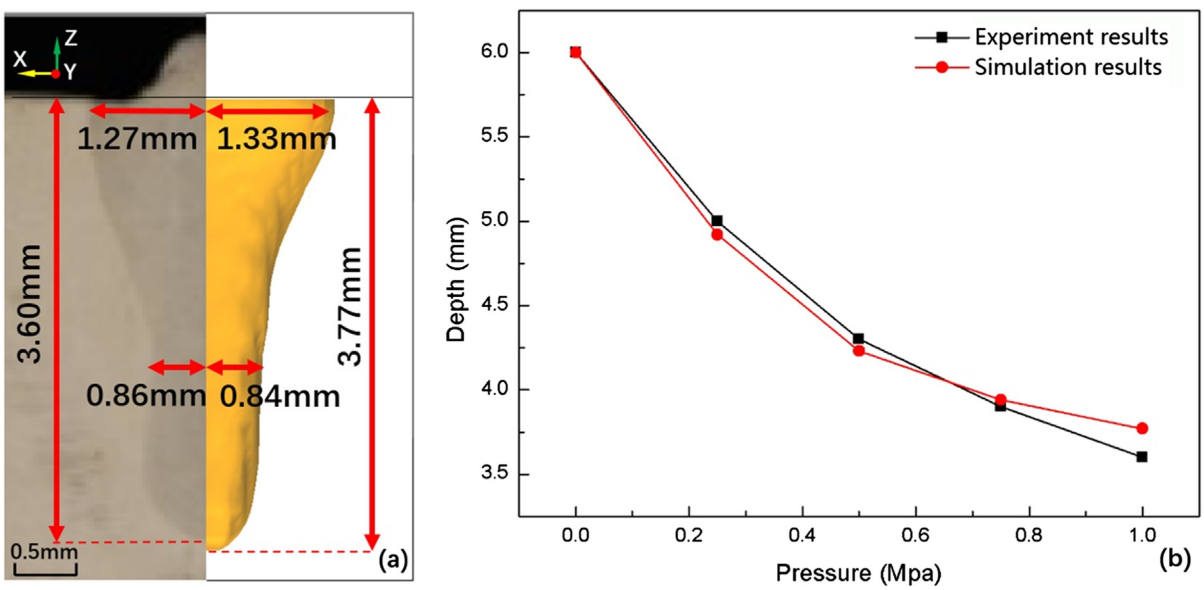

Fig. 4. The comparisons of the experimental results and predicted. (a) Weld profile comparisons between experiment [38] and simulation (Process No. 6); (b) Weld depth comparisons between experiment and simulation (Process No. 2, 3, 4, 5 and 6).

simply. In addition, to investigate the effect of deep water environment on the keyhole and weld pool dynamics, we modelled the process under the condition of 5.0 MPa hydrostatic pressure with the same laser energy absorption in Process No. 6. That penetration depth was 0.98 mm. It can be found that the depth of weld decreased with the increase of water pressure. In addition, the decrement was smaller gradually. Similar findings can also be found in other studies [41]. In a word, the experimental results and the independent literature results above indicate that the underwater pressure environment has a significant effect on the weld. The morphology of the weld underwater was similar to that in atmosphere environment, but the depth of the weld underwater decreases by using the same welding parameters.

3.2. Mesoscopic physical process of underwater laser welding

To understand the effect of water on laser welding, we investigated the mesoscopic physics behaviors of keyhole and weld pool dynamics in two typical laser welding cases (Process No.1 and Process No. 2) using numerical simulations. Fig. 5 shows the transient keyhole evolutions during underwater laser welding (0.1 MPa ambient pressure). Fig. 5a shows the three dimensional view of the predicted result. For convenience, the longitudinal section (the ZOY plane) of the weld bead and the three dimensional keyhole surface (the conical surface in Fig. 5) were extracted to analyze. The color represents the temperature. The white color represents 300 K. The temperature at green area is larger than 1800 K. The green area following keyhole is weld pool. In Fig. 5, the keyhole profile was irregular and dynamically evolving in the welding process. The purple zones were the regions directly irradiated by the laser beam, where the temperature was higher than the other place. In the initial state of 7.70 ms, the keyhole is formed (Fig. 5b). The temperature of the whole keyhole surface was nearly over 3200 K. The morphology of the keyhole was conical. The wall of the keyhole hardly oscillated. Many irregular humps (sized in tens of microns) occurred on the keyhole as the keyhole was deeper and deeper. The breakup of the keyhole can be determined in Fig. 5c. At 214.63 ms, the plate was full penetrated (Fig. 5d). The keyhole was in a quasi-steady state. A large uncooled weld pool formed behind. In the state of 287.08 ms, the depth of the keyhole did not increase compared with the that in Fig. 5e. The highest temperature was around 3500 K. In addition, many irregular humps occurred on the keyhole. The temperature at the contraction of the keyhole was as high as 3500 K, but the temperature at the opening of the keyhole was only 1900 K (Fig. 5f).

These results indicate that the dynamical behaviors of the keyhole in the water are similar to that in the atmosphere. The temperature distribution of the keyhole surface is uneven. In addition, the keyhole oscillation is existed. The breakup of the keyhole cause by oscillation can be observed.

Moreover, the transient weld pool dynamics in the water (about 0.1 MPa) was studied, as shown in Fig. 6. The arrow denotes the velocity of the molten metal, the direction of the arrow represents the direction of the velocity, and the color of the arrow indicates the value. The black line is a temperature isoline of solidifying point which was the interface between the solid and liquid regions. The region bounded by the line can be considered as the weld pool. The flow in the pool was very complex, and vortexes with different

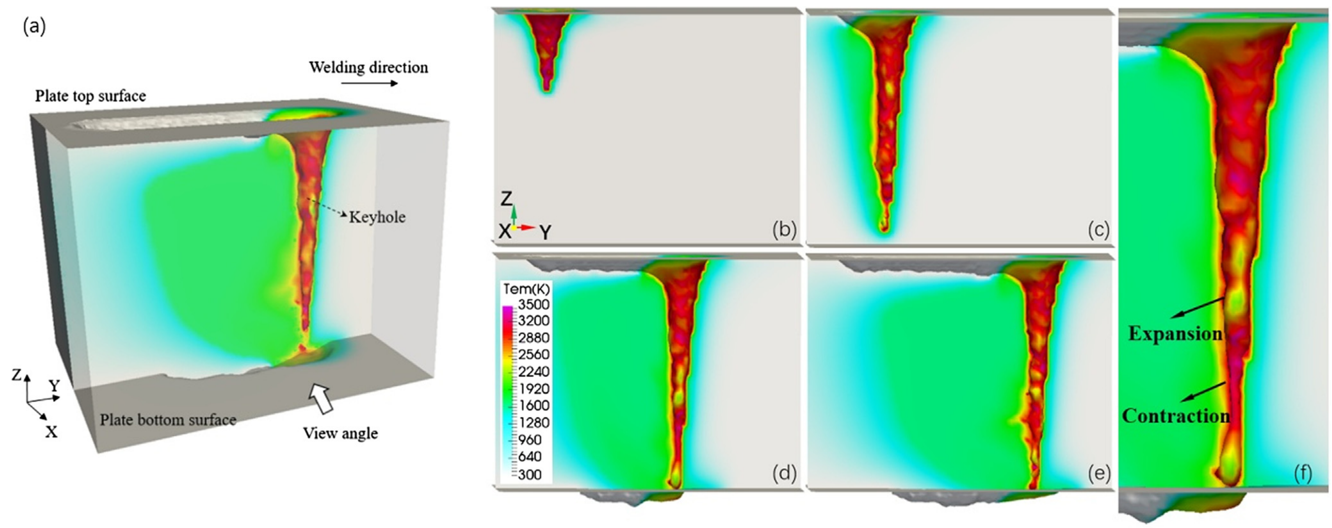

Fig. 5. The transient keyhole evolutions under 0.1 MPa pressure. (a) 3d view of the result, (b) 7.70 ms, (c) 58.88 ms, (d) 214.63 ms, (e) 287.08 ms, (f) local enlarged view of (d).

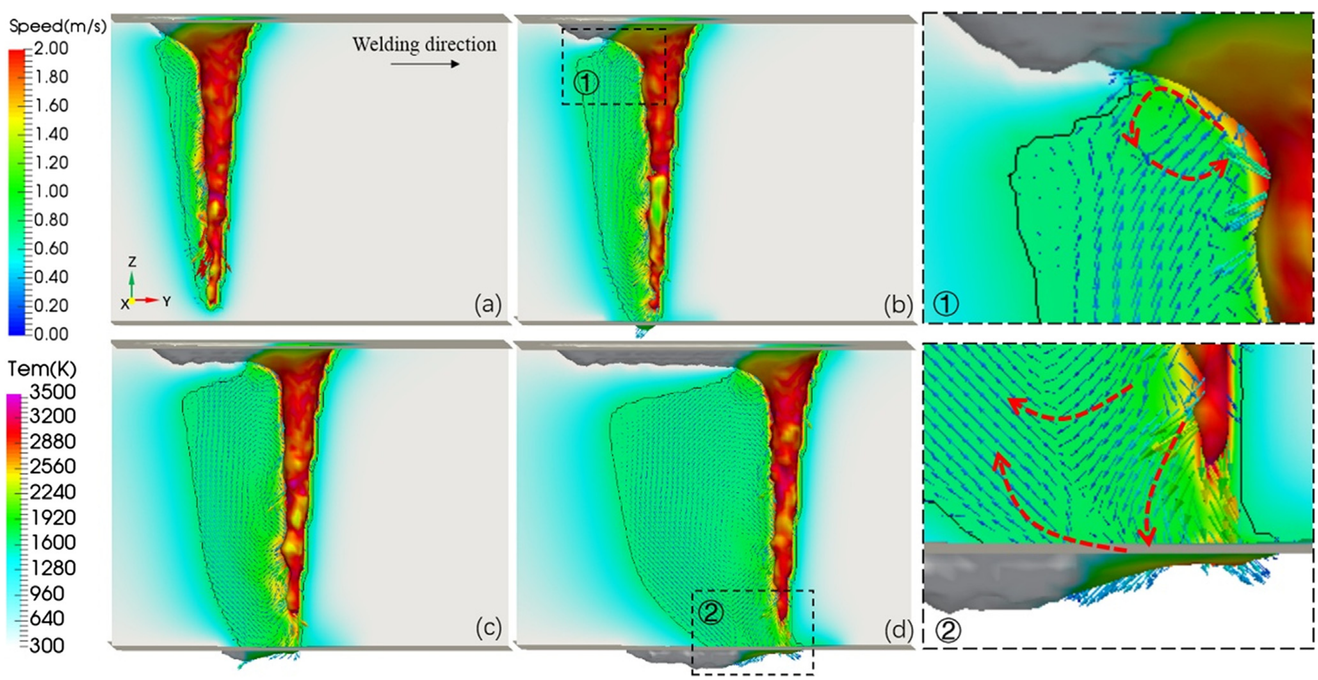

Fig. 6. Predicted fluid flow in the weld pool during underwater laser welding. (a) 59.69 ms, (b) 99.73 ms, (c) 150.89 ms, (d) 251.58 ms.

rotational directions can be observed. At 59.69 ms, the keyhole was about to penetrate the plate (Fig. 6a). It was deep and thin with a narrow weld pool behind it. The wall of the keyhole was uneven and oscillating at all times which caused the synchronous fluid flows in the pool near the irregular wall. The laser beams directly irradiated the contraction region of the keyhole surface, which made the region had a tendency to expand at the next moment. Meanwhile, the expansion of the keyhole would drive the nearby molten metal to flow at a downward high velocity (the peaks were greater than 2 m/s). As the keyhole depth increased, the plate was penetrated (Fig. 6b). Root humping formed at the lower surface of the plate. The weld pool behind the keyhole expanded significantly. The weld pool in front of the keyhole was kept as a liquid film. As shown in enlarged partial view of Fig. 6b (①), a counter-clockwise vortex can be observed in the upper part of the pool behind the keyhole. The vortex was caused by the temperature gradient which flowed from the high temperature zone to the low temperature zone. In the state of 150.89 ms, the weld pool behind the keyhole was further expanded (Fig. 6c). A number of vortexes, mostly clockwise, caused by oscillations on the wall of keyhole can be observed in the molten metal. When the weld pool behind the keyhole became wider at 251.58 ms, the fluid flow around the keyhole, especially near the upper and lower part of the keyhole, was very complicated (Fig. 6d), as shown in enlarged partial view of Fig. 6d (②). The downward velocity of the liquid can be observed at the bottom of the weld pool, which may cause the liquid to leave the pool from the bottom. And the velocity to the rear was the cause of the forming of root humping. However, as a whole, the fluid flow seemed to be uniform. This indicated that the flow in the weld pool was still mainly caused by recoil pressure. From above, it was known that the flow in the weld pool was complex but followed basic rules. The flow caused by temperature gradient was still existed but recoil pressure was the main factor leading flow in the weld pool.

In order to better understand the influence of water, we compared the flow characteristics of weld pool between welding processes in water and in air (about 0.1 MPa), as shown in Fig. 7. In both water and air, the highest velocity of the flow in the pool reached 2 m/s. Obviously, the morphology of the weld pool in the water (Fig. 7b) was significantly different from that under atmosphere pressure (Fig. 7a). The weld pool in the atmosphere was wider at the top surface and the bottom surface of the plate. The top and bottom surfaces of the weld pool contacted with the water were greatly reduced. The diameter of the pool contacted with the water was only about twice of the diameter of the keyhole. It was caused by the cooling effect of water on the surface of the weld pool. Due to the strong convection heat transfer of water on the surface of the weld pool, the solidification of the molten metal contacted with water is faster, and the unique morphology of the weld pool is formed.

Another particular phenomenon was that the flow patterns in the weld pool had changed. When the plate was in the atmosphere, the counterclockwise vortex in the upper part of the pool was violent. However, the vortex upper was confined by surface solidification in water, and concentrated in a small region. These results indicate that water has a great influence on the morphology of weld pool and the flow in it. The cooling effect of water will reduce the opening of the pool and form a unique morphology of the pool. It will also change the flow direction in the weld pool.

3.3. Effect of the water depth on the keyhole and weld pool dynamical behaviors.

To further figure out the effect of the water depth on the dynamics, the underwater welding processed under different pressure were investigated. We compared the mesoscopic dynamics of the keyhole and weld pool in three typical laser welding cases (Process No. 2 in Table 3: Underwater, Pressure: 0.1 MPa; Process No.6 in Table 3: Underwater, Pressure: 1 MPa; Process No.7 in Table 3: Underwater, Pressure: 5 MPa).

Fig. 8 showed the comparison of the profiles of the keyhole under 0.1 MPa (shallow water), 1 MPa (underwater 100 m) and 5 MPa (underwater 500 m) pressure. In order to compare the characteristics of the keyhole and the weld pool under different ambient pressures clearly, the same color bar was adopted. From Fig. 8, we can find two important phenomena. First, when the water pressure was increased, the average temperature of the keyhole surface was increased. As shown in Fig. 8a, the temperature of most of the keyhole surface under 0.1 MPa was about 3200 K. The highest surface temperature of the keyhole under 1 MPa was above 3660 K. When the ambition pressure increased to 5 MPa, the highest temperature was up to 4500 K. The increase of the temperature was because the boiling point was increased and the laser energy was enough to heat the surface to the boiling point. It indicates that the water depth can greatly affect the dynamics of the keyhole and the weld pool.

In addition, we observed that the behaviors of the keyhole and weld pool were also changed. The morphology of the keyholes was conical. And with the increase of pressure, the depth of the keyhole decreased. The number of humps on the surface of the keyhole decreased, and the oscillations of keyhole were also weakened. It is related to the energy distribution under different ambient pressures. More power per unit depth of keyhole is necessary when the boiling point is increased. It will hinder the effect of punching the fluid near the keyhole bottom by recoil pressure. Then the keyhole depth is decreased. As a result, the shallow keyhole is hard to be broken and the keyhole oscillation is weakened.

Fig. 9 showed the comparison of the fluid flow patterns in the weld pool under 1 MPa and 5 MPa pressure. From Fig. 9, it was clear that with the increase of pressure, the depth of the keyhole decreased, the temperature of the keyhole surface increased, and the shape of the weld pool changed, respectively. The thickness and the temperature gradient of the weld pool ahead of the keyhole were increased due to the temperature of the keyhole surface was higher under 5 MPa pressure. The flow of the weld pool behind the keyhole was affected mainly by the oscillation of the keyhole surface. Concretely, the keyhole surface was irradiated by laser, resulting in intense evaporation. The keyhole was subjected to the recoil pressure perpendicular to the surface (main direction downward), leading the weld pool near the middle and lower part of the keyhole flowed downward to the bottom of the keyhole. The liquid metal was confined by the solid–liquid interface and flowed to the back of the weld pool along the interface to form a clockwise vortex (Fig. 9 ①). The surface of the weld pool solidified rapidly

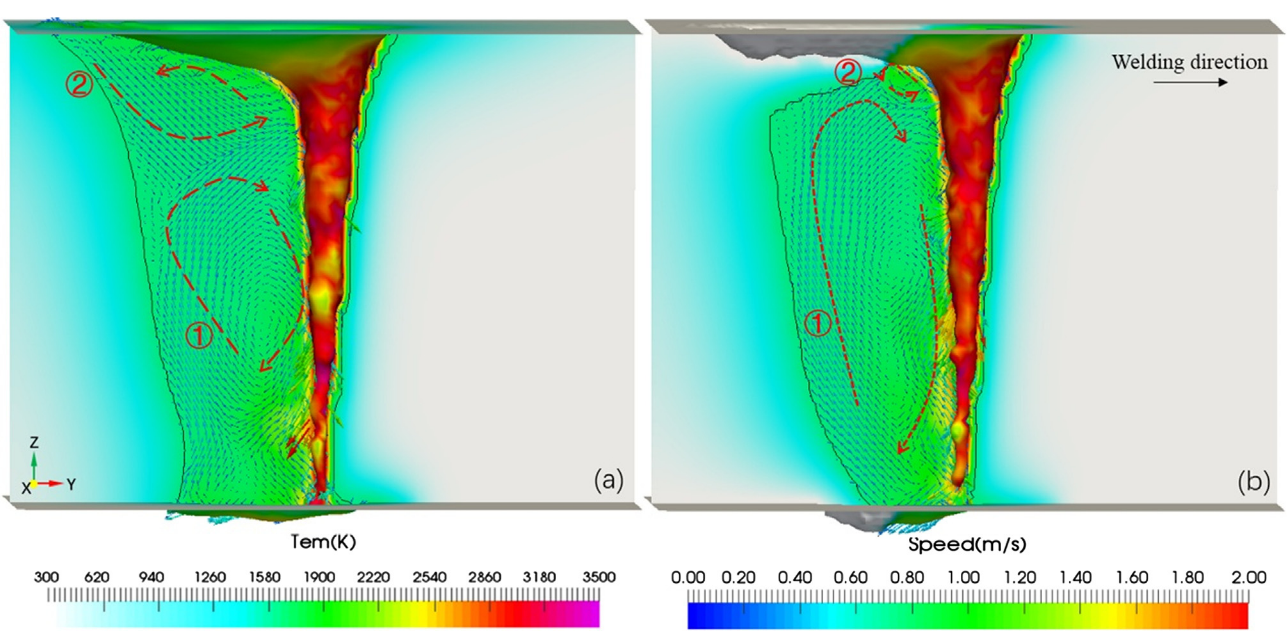

Fig. 7. Comparison of the fluid flow in the weld pool under different environments (167.92 ms). (a) In the atmosphere; (b) In the water.

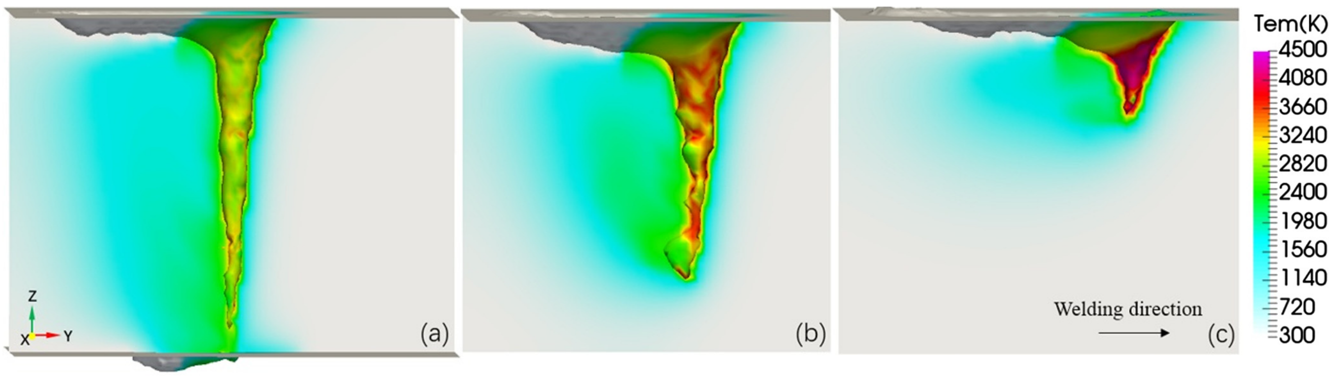

Fig. 8. The predicted keyholes in different water depths (185.45 ms). (a) 0.1 MPa, (b) 1 MPa, (c) 5 MPa.

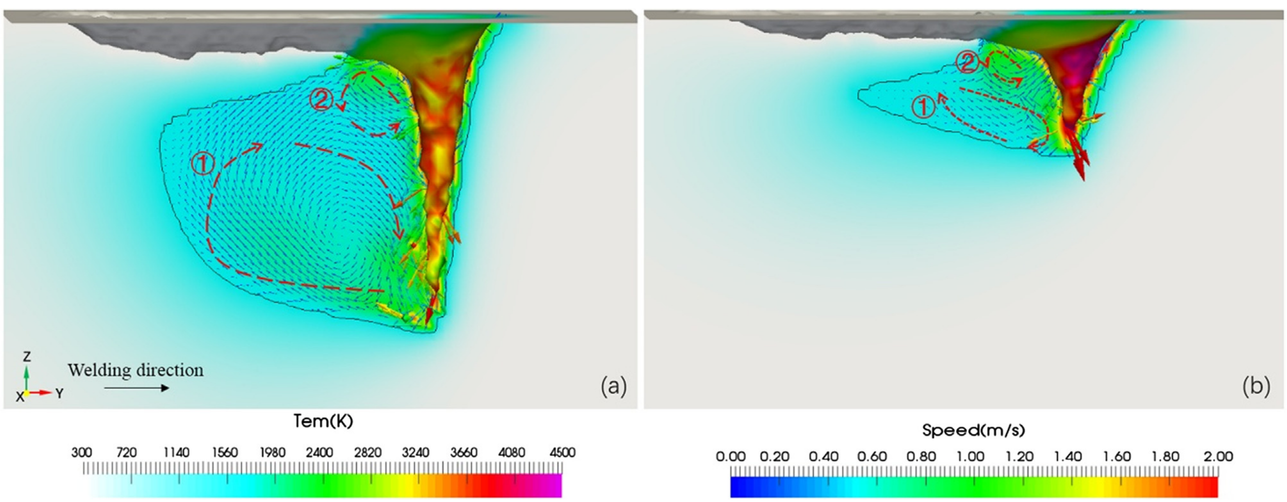

Fig. 9. Comparison of the fluid flow in the weld pool under different pressures in the water (246 ms). (a) 1 MPa; (b) 5 MPa.

due to the effect of water cooling. The weld pool below the solidified part, which did not have time to solidify in such a short period of time, remained liquid. However, due to bounded by the solidified part of the surface, the flow in the weld pool was different from that under atmosphere pressure. The counterclockwise vortex in the upper part of the molten pool caused by the temperature gradient induced Marangoni force was concentrated in the small unsolidified region, near the opening of the keyhole (Fig. 9 ②). As a result, the flow effect near the opening of the keyhole was weakened. From above, the temperature of the keyhole and weld pool in deep water was obviously increased and the fluid flow was significantly influenced due the changes of the weld pool morphology.

Through the above analysis, we found that the cooling effect of water on the workpiece surface existed in the underwater environment, and the water pressure can obviously change the boiling point of the material, which was quite different from laser welding in atmosphere. As a result, a shallower but more stable keyhole is formed in underwater laser welding process under 5 MPa pressure. This study can provide better understanding of the heat and mass transfer behaviors in underwater laser welding.

4. Conclusions

In this paper, we investigated the dynamical behaviors of the keyhole and weld pool in underwater laser welding process. A clear physical picture of how the water influenced the laser welding process was present. The major findings of this paper were as follows:

(1) A novel mathematical model of laser welding process in deep water environment was developed. The predicted results of underwater laser welding of stainless steel in different depths were consistent with experimental results.
(2) The dynamical behaviors of the keyhole in the water were similar to that under atmosphere pressure. The temperature distribution of the keyhole surface was uneven. In addition, the keyhole oscillation occurred.
(3) Strong water cooling effect on the surface of weld pool made the morphology of the weld pool different from that in the atmosphere. As a result, the flow patterns were changed significantly.
(4) When the water pressure increases, the surface temperature of the keyhole increases. A shallower but more stable keyhole is formed in underwater laser welding process under 5 MPa pressure.

Conflict of interest

The authors declare that they have no conflict of interest.

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References

[1] M. Yoda, M. Tamura, T. Fukuda, K. Shiihara, K. Sudo, T. Maehara, Y. Morishima, H. Kato, H. Ichikawa, Underwater laser beam welding for nuclear reactors, in: 20th International Conference on Nuclear Engineering and the ASME 2012 Power Conference, American Society of Mechanical Engineers, 2012, pp. 191–195.
[2] H. Li, D. Liu, Y. Yan, N. Guo, J. Feng, Microstructural characteristics and mechanical properties of underwater wet flux-cored wire welded 316L stainless steel joints, J. Mater. Process. Technol. 238 (2016) 423–430.
[3] N. Guo, X. Xing, H. Zhao, C. Tan, J. Feng, Z. Deng, Effect of water depth on weld quality and welding process in underwater fiber laser welding, Mater. Des. 115 (2017) 112–120.
[4] Y. Ogawa, T. Irie, Y. Ono, H. Matsushita, M. Sato, Effect of shielding gas on underwater CO_2 welding, Proceedings of the 17th OMAE, 1998.
[5] J. Wallace, MATERIALS PROCESSING: 100 kW fiber laser, power meter serve industry, Laser Focus World 49 (12) (2013) 13–14.
[6] X. Shi, J. Wang, Y. Feng, X. Zhang, C. Wang, T. Bo, H. Liu, Study on the effect of fiber refractive index distribution on the homogenization of semiconductor laser beam. In Advanced Laser Technology and, Appl. Int. Soc. Opt. Photonic. 10844 (2018) 108440U.
[7] B. McAllister, Robust focusing optics for high-power laser welding, in high-power laser materials processing: lasers, beam delivery, diagnostics, and applications III, Int. Soc. Opt. Photonic. 8963 (2014) 89630A.
[8] X. Zhang, W. Chen, E. Ashida, F. Matsuda, Relationship between weld quality and optical emissions in underwater Nd: YAG laser welding, Opt. Laser Eng. 41 (5) (2004) 717–730.
[9] I. Morita, K. Owaki, H. Yamaoka, C.C. Kim, Study of underwater laser welding repair technology, Weld World 50 (7–8) (2006) 37–43.
[10] X. Zhang, E. Ashida, S. Shono, F. Matsuda, Effect of shielding conditions of local dry cavity on weld quality in underwater Nd: YAG laser welding, J. Mater. Process. Technol. 174 (1–3) (2006) 34–41.
[11] Z. Huang, Z. Luo, S. Ao, Y. Cai, Underwater laser weld bowing distortion behavior and mechanism of thin 304 stainless steel plates, Opt. Laser Technol. 106 (2018) 123–135.
[12] J. Ning, L. Zhang, Q. Bai, X. Yin, J. Niu, J. Zhang, Comparison of the microstructure and mechanical performance of 2a97 al-li alloy joints between autogenous and non-autogenous laser welding, Mater. Des. 120 (2017) 144–156.
[13] G. Sepold, K. Teske, Results of underwater welding with high power CO_2 lasers, International Congress on Application of Laser and Electro-Optics (ICALEO) 38 (1983) 87–89.
[14] G.J. Shannon, J. Watson, W.F. Deans, Investigation into the underwater laser welding of steel, J. Laser Appl. 6 (4) (1994) 223–229.
[15] L. Lindhorst, O. Mahrenholtz, On the influence of wet underwater welding on CTOD-δ5, Int. J. Fract. 97 (1–4) (1999) 263–277.
[16] X. Wen, G. Jin, X. Cui, X. Feng, B. Lu, Z. Cai, Y. Fang, Underwater wet laser cladding on 316L stainless steel: A protective material assisted method, Opt. Laser Technol. 111 (2019) 814–824.
[17] G.J. Shannon, W. McNaught, W.F. Deans, J. Watson, High power laser welding in hyperbaric gas and water environments, J. Laser Appl. 9 (3) (2012) 129.
[18] E.H. Amara, R. Fabbro, Modelling of gas jet effect on the melt pool movements during deep penetration laser welding, J. Phys. D Appl. Phys. 41 (5) (2008) 055503.
[19] N. Guo, Y. Fu, X. Xing, Y. Liu, S. Zhao, J. Feng, Underwater local dry cavity laser welding of 304 stainless steel, J. Mater. Process, Technol, 2018.
[20] A. Anand, A. Khajuria, Welding processes in marine application: A review, Int. J. Mech. Eng. Rob. Res. 2 (2015) 215–225.
[21] L. Liu, H. Wang, B. Yu, Y. Xu, J. Shen, Improved algorithm of light scattering by a coated sphere, China Particuol. 5 (3) (2007) 230–236.
[22] R.G. Grainger, J. Lucas, G.E. Thomas, G.B.L. Ewen, Calculation of Mie derivatives, Appl. Opt. 43 (28) (2004) 5386–5393.
[23] K.E. Thelning, Steel and its heat treatment, Butterworth-heinemann, 2013.
[24] G.E. Totten (Ed.), Handbook of residual stress and deformation of steel, ASM international, 2002.
[25] B. Huang, X. Chen, S. Pang, Z. Ren, A three-dimensional model of coupling dynamics of keyhole and weld pool during electron beam welding, Int. J. Heat Mass Transf. 115PB (2017) 159–173.
[26] J. Eggers, Nonlinear dynamics and breakup of free-surface flows, Rev. Mod. Phys. 69 (3) (1997) 865.
[27] R. Rai, T. DebRoy, Tailoring weld geometry during keyhole mode laser welding using a genetic algorithm and a heat transfer model, J. Phys. D Appl. Phys. 39 (2006) 1257–1266.
[28] J. Zhou, H.L. Tsai, T. Lehnhoff, Investigation of transport phenomena and defect formation in pulsed laser keyhole welding of zinc-coated steels, J. Phys. D Appl. Phys. 39 (2006) 5338–5355.
[29] S. Pang, X. Chen, J. Zhou, X. Shao, S. Gong, J. Xiao, Dynamics of vapor plume in transient keyhole during laser welding of stainless steel: local evaporation, plume swing and gas entrapment into porosity, Opt. Lasers Eng. 82 (2016) 28–40.
[30] S. Pang, X. Chen, J. Zhou, X. Shao, C. Wang, 3D transient multiphase model for keyhole, vapor plume, and weld pool dynamics in laser welding including the ambient pressure effect, Opt. Lasers Eng. 74 (2015) 47–58.
[31] S. Pang, L. Chen, J. Zhou, Y. Yin, T. Chen, A three-dimensional sharp interface model for self-consistent keyhole and weld pool dynamics in deep penetration laser welding, J. Phys. D Appl. Phys. 44 (2) (2011) 025301.
[32] Q. Tang, S. Pang, B. Chen, H. Suo, J. Zhou, A three dimensional transient model for heat transfer and fluid flow of weld pool during electron beam freeform fabrication of Ti-6-Al-4-V alloy, Int. J. Heat Mass Transf. 78 (7) (2014) 203–215.
[33] S. Pang, K. Hirano, R. Fabbro, T. Jiang, Explanation of penetration depth variation during laser welding under variable ambient pressure, J. Laser Appl. 27 (2) (2015) 022007.
[34] R. Fabbro, K. Hirano, S. Pang, Analysis of the physical processes occurring during deep penetration laser welding under reduced pressure, J. Laser Appl. 28 (2) (2016) 022427.
[35] K. Hirano, R. Fabbro, M. Muller, Experimental determination of temperature threshold for melt surface deformation during laser interaction on iron at atmospheric pressure, J. Phys. D Appl. Phys. 44 (2011) 435402.
[36] S. Osher, R. Fedkiw, Level set methods and dynamic implicit surfaces, Springer, 2003.
[37] A.J. Chorin, Numerical solution of the Navier-Stokes equations, Math. Comp. 22 (104) (1968) 745–762.
[38] H. Futoshi, K. Yousuke, M. Masami, I. Hiroshige, K. Katsuyoshi, Full penetration characteristics of laser welding on stainless steel using high power and a brightness laser, J. Japan Welding Soc. 36 (1) (2018) 122–128.
[39] E. Assuncao, S. Williams, D. Yapp, Interaction time and beam diameter effects on the conduction mode limit, Opt. Laser Eng. 50 (6) (2012) 823–828.
[40] H. Zhao, T. DebRoy, Macroporosity free aluminum alloy weldments through numerical simulation of keyhole mode laser welding, J. Appl. Phys. 93 (12) (2003) 10089–10096.
[41] H. Shen, D. Feng, Z. Yao, Modeling of underwater laser drilling of alumina, J. Manuf. Sci. Eng. 139 (4) (2017) 041008.